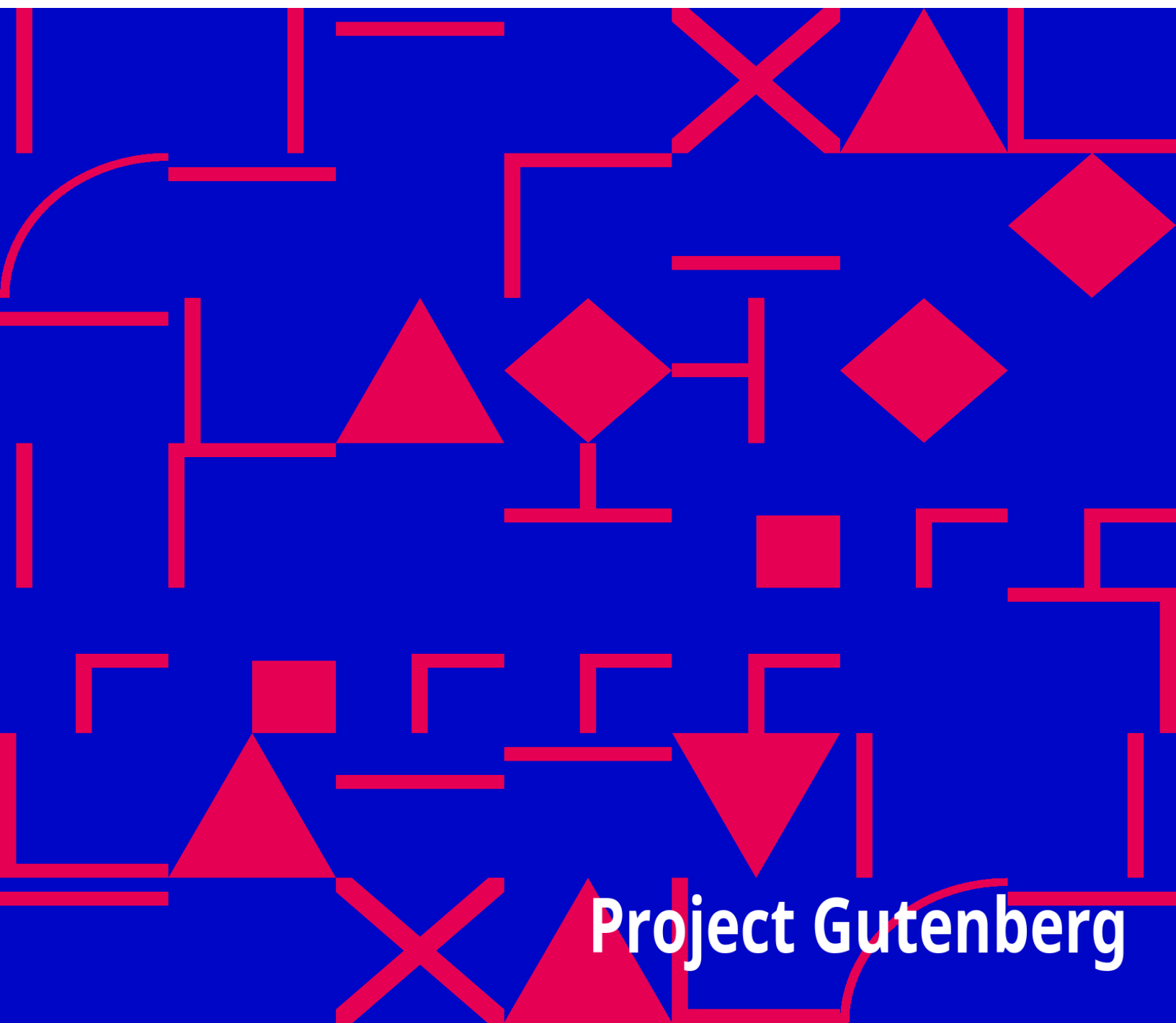


The Value of Zeta(3) to 1,000,000 places

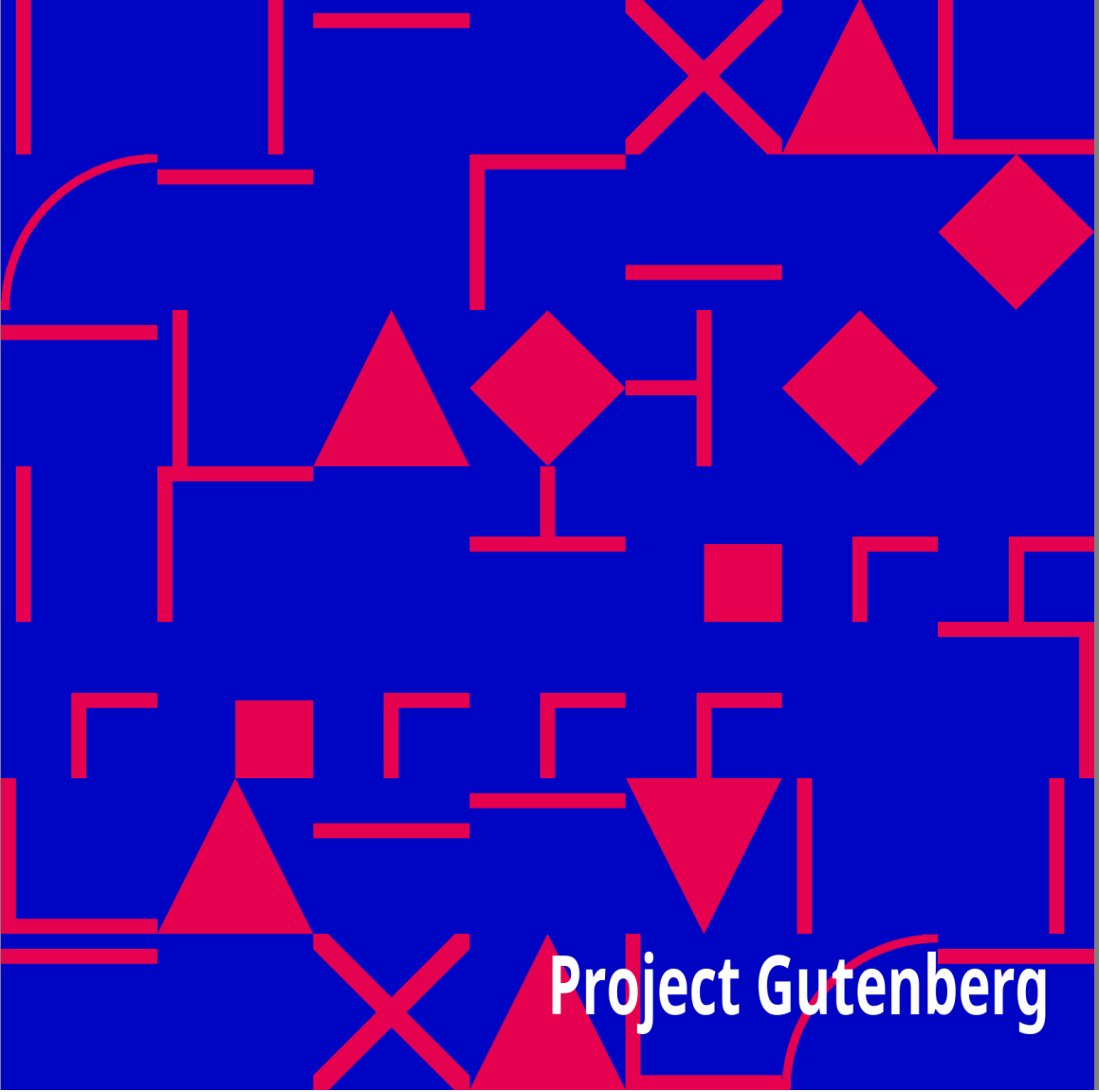
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The lower half of the page features a complex, abstract pattern of red geometric shapes on a blue background. The shapes include various sizes of triangles, diamonds, squares, and lines, some of which are arranged to form larger, more complex structures like a large 'X' and a large 'L'. The pattern is dense and covers the entire lower half of the page.

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Mathematical constants and numbers

edited by Simon Plouffe Associate Professor LaCIM, University of Quebec at Montreal <http://www.lacim.uqam.ca/pi> : Plouffe's Inverter
 plouffe@math.uqam.ca

The value of Zeta(3) to 1,000,000 decimal digits. the number is defined as $\sum(1/n^3, n=1..infinity)$, the sum of inverses of cubes and equals 1.2020569031...

Computed by : Sebastian Wedeniwski (wedeniws@de.ibm.com) who computed more than 128 million digits using this more efficient formula found by Theodor Amdeberhan and Doron Zeilberger.

$$Zeta(3) = \frac{1}{24} \sum_{n=0}^{\infty} \frac{(-1)^n}{3^{n+1}} \frac{A(n)}{(2n+1)! (2n)! n!} \frac{(3n+2)! ((4n+3)!)}{(n!)^3}$$

with $A(n) := 126392 n^5 + 412708 n^4 + 531578 n^3 + 336367 n^2 + 104000 n + 12463$ given by Theodor Amdeberhan and Doron Zeilberger (see [1]).

References: =====

[1] T. Amdeberhan und D. Zeilberger: Hypergeometric Series Acceleration via the WZ Method, Electronic Journal of Combinatorics (Wilf Festschrift Volume) 4 (1997).

[2] B. Haible, T. Papanikolaou: Fast multiprecision evaluation of series of rational numbers, Technical Report TI-97-7, Darmstadt University of Technology, April 1997.

[3] S. Wedeniwski: Piologie - Eine exakte arithmetische Bibliothek in C++, Technical Report WSI 96-35, Tuebingen University, available by anonymous ftp from "<ftp://ftp.informatik.uni-tuebingen.de/pub/CA/software/Piologie/>" or "<ftp://ruediger.informatik.uni-tuebingen.de/Piologie/>".

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